

Oct 8 notes

$$a_1 = 0, \quad a_{n+2} = \sqrt{6 + a_n}. \quad \text{Find upper bound}$$

$$a_2 = \sqrt{6} \quad a_3 = \sqrt{6 + \sqrt{6}} \quad a_4 = \sqrt{6 + \sqrt{6 + \sqrt{6}}}$$

Suppose $a_n \geq a_1$ and $a_1 = 0$

$$a_1 < 3 \quad \rightarrow \quad a_n \leq 6 \quad \text{for } n \geq 2$$

$$a_k \leq 3$$

$$a_k + 6 \leq 9 \quad \rightarrow \quad \sqrt{a_k + 6} \leq \sqrt{9} = \sqrt{a_k + 6} \leq 3 \quad \text{true}$$

Bounded above by 3